



Effect of early and adequate empiric antibiotic therapy for hospital-acquired sepsis in the age of antimicrobial resistance: a target trial emulation using machine learning

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INTRODUCTION

BACKGROUND

- Antimicrobial resistance (AMR) is a leading public health concern, particularly in the context of hospital-acquired infections.
- Delays in appropriate antibiotics increase morbidity and mortality, however studies assessing burden and prognostic impact of antibiotic therapy in this critically ill population at the hourly level are limited.
- Given the limitations of randomized trials, innovative strategies using real-world data and computational methods are needed.

OBJECTIVE:

- To leverage a combination of target trial emulation and machine learning to estimate causal effects of early and adequate antibiotic therapy in hospital-onset sepsis, as well as the moderating role of AMR

METHODS

STUDY DESIGN

- Retrospective cohort study designed with the **target trial emulation framework** using data from UF Integrated Data Repository (IRB201900652).
- Approximately 140,000 hospital encounters with documented bacterial infection from 2010 to 2020 were examined from eligibility for adult patients who were hospitalized hospital-onset sepsis. (**Figure 1**)

Target Trial Specification

Eligibility Criteria	→	Adult patients hospitalized for at least 48 hours with clinical suspicion of new onset of sepsis based on US CDC Definitions
Treatment Groups	→	Receipt of antibiotic regimen within 6, 12, 18, 24, 36 or 48 hours of blood culture collection that: 1. Covers the bacterial pathogen via AST* 2. Does not cover bacterial pathogen via AST*
Assignment/Randomization	→	Randomization emulated using baseline and time-varying covariates within 6, 12, 18, 24, 36 or 48 of blood culture collection
Follow-Up Period	→	Blood culture collection → 30 days or hospital discharge, or death
Outcome	→	In-hospital mortality within 30 days
Causal Contrast	→	Observational analog Per-protocol effect

*Antibiotic Susceptibility Testing

STATISTICAL APPROACH

- Features included were patient characteristics, pathogen-specific risk factors, and clinical severity metrics. (**Table 1, Figure 2**)
- Propensity scores estimated using machine learning methods including logistic regression, lasso regression, gradient-boosted decision trees, random forest, and ensemble learning. Inverse-Probability Weights (IPW) were calculated from most optimal model. (**Table 2**)
- Marginal effects estimated via marginal structural Cox proportional hazards models in three methods: unadjusted, IPW, doubly robust (**Figure 3, Table 3**)

RESULTS: POPULATION SUMMARY

Figure 1: Flowchart of study inclusion

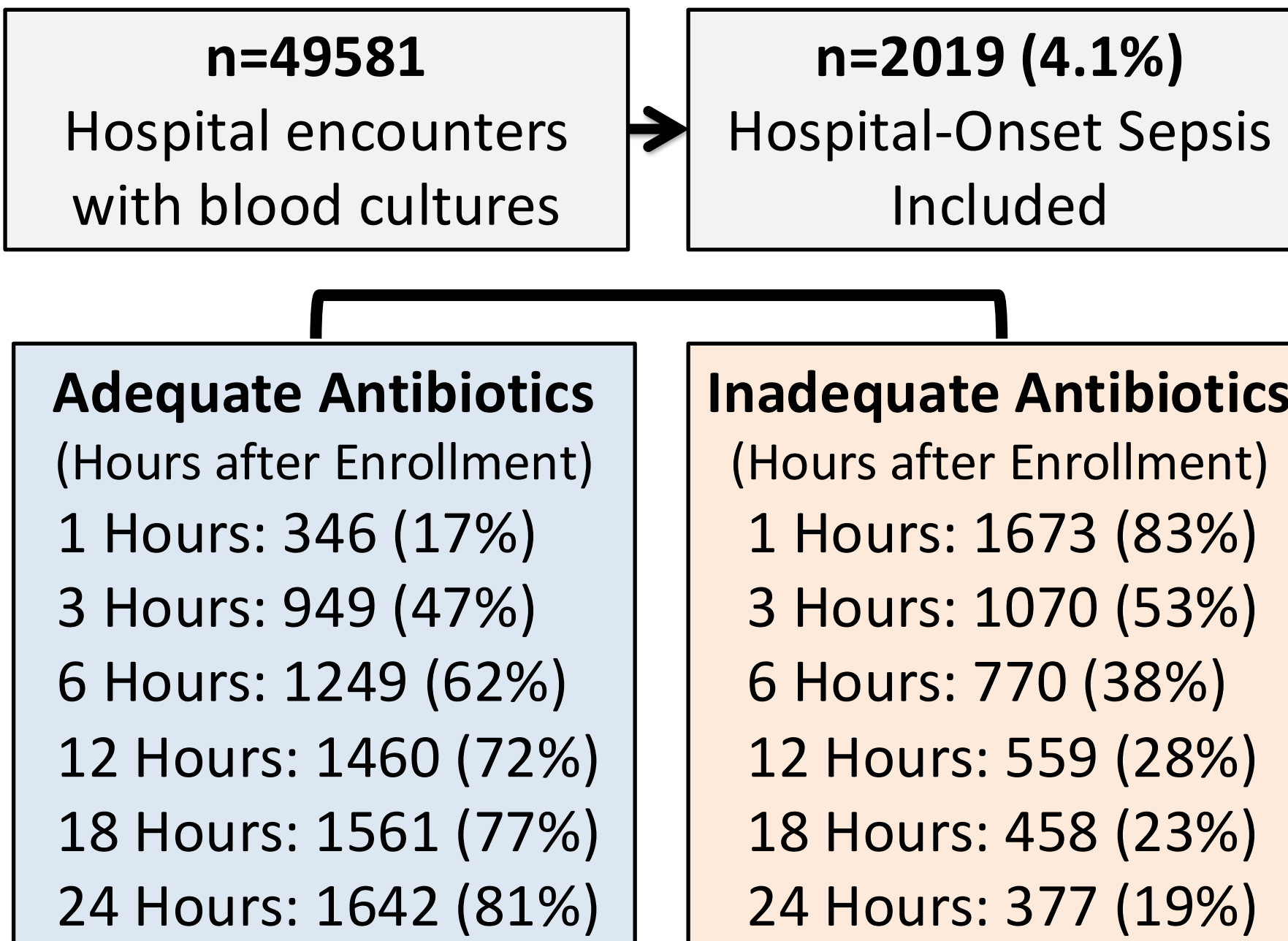


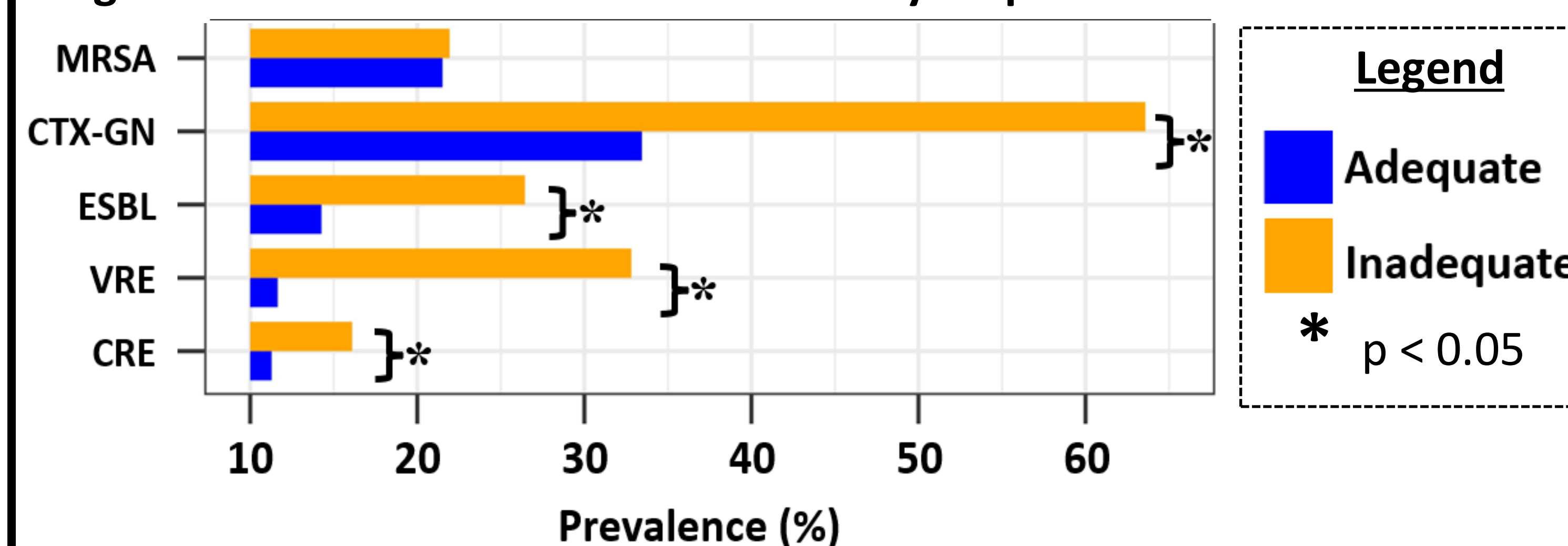
Table 2: Covariate Balance (ASMD <0.05) by method

Method	Covariates Balanced (%)
Unadjusted	53/77 (69%)
LASSO	75/77 (97%)
Logistic-Trim	77/77 (100%)
Gradient Boost	56/77 (72%)
Random Forest	55/77 (71%)
Ensemble Learning	61/77 (79%)

Table 1: Descriptive Summary by Treatment Group

Color of P-value indicates direction of effect	Antibiotics at 24 Hours		P Value
	Adequate (n=1642)	Inadequate (n=377)	
Demographics:			
Age, years ± SD	65.4 ± 16.5	65.1 ± 15.6	0.747
Sex, Female	714 (43.5)	186 (49.3)	0.045
Race, Black	401 (24.4)	107 (28.4)	0.134
Past Med. History:			
Chronic Pulmonary	652 (39.7)	172 (45.6)	0.040
Cerebrovascular	387 (23.6)	108 (28.6)	0.045
Hemiplegia/Paraplegia	107 (6.5)	40 (10.6)	0.008
Clinical Factors			
Vasopressor Initiation	982 (59.8)	193 (51.2)	0.003
Source of Infection			
Intra-Abdominal	313 (19.1)	97 (25.7)	0.005
Bone/Joint	113 (6.9)	16 (4.2)	0.076
Genitourinary	400 (24.4)	133 (35.3)	<0.001
Other	855 (52.1)	224 (59.4)	0.012
Antimicrobials			
Vancomycin	1283 (78.1)	204 (54.1)	<0.001
Cefepime	1063 (64.7)	196 (52)	<0.001
Metronidazole	530 (32.3)	116 (30.8)	0.614
Piperacillin-Tazobactam	371 (22.6)	64 (17)	0.020
Ceftriaxone	162 (9.9)	58 (15.4)	0.003
Amikacin	70 (4.3)	3 (0.8)	0.002

Figure 2: Prevalence of resistance class by empiric antibiotics at 24 hours



RESULTS: CAUSAL MODEL

Figure 3: Mortality probability curves by treatment groups at 24 hours for unadjusted and IPW-adjusted approaches

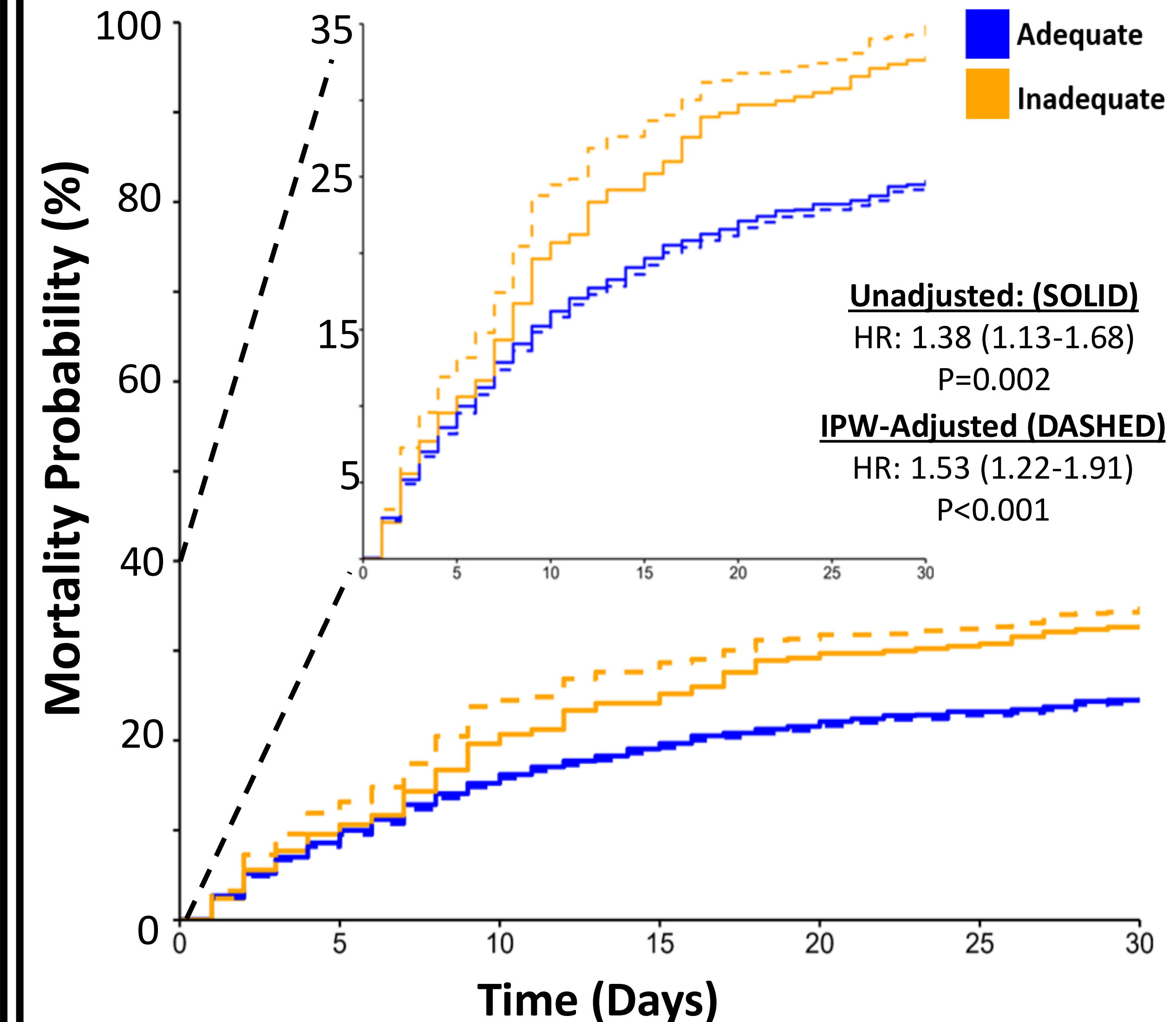


Table 3: Average treatment effects of inadequate empiric antibiotic therapy for each time after enrollment.

Time from Enrollment	Unadjusted HR (95%CI)	IPW-Adjusted HR (95%CI)
1 Hours	1.13 (0.89-1.42)	1.07 (0.83-1.38)
3 Hours	1.10 (0.93-1.31)	1.18 (1.01-1.41)
12 Hours	1.13 (0.94-1.36)	1.29 (1.09-1.57)
24 Hours	1.38 (1.13-1.68)	1.53 (1.22-1.91)
P for Trend (24 Hours)	P=0.011	P=0.002

CONCLUSION

- Approximately one in five patients with hospital-onset sepsis received inadequate empiric therapy, largely due to antimicrobial resistance
- Delays in adequate treatment starting at 3 hours after culture collection caused increasing mortality
- We highlight the value of integrating real-world data sources, such as electronic health records and antibiotic susceptibility testing, with expert-driven modeling and machine learning for causal effect estimation in complex clinical scenarios